

Final Paper

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Introduction

(JUNE) - Establish context for study: - Relevance to NCOS - Relevance to other ecosystems - Synthesize main findings from other papers (connect ideas, don't just list) - Articulate main question(s) for the study

Methods

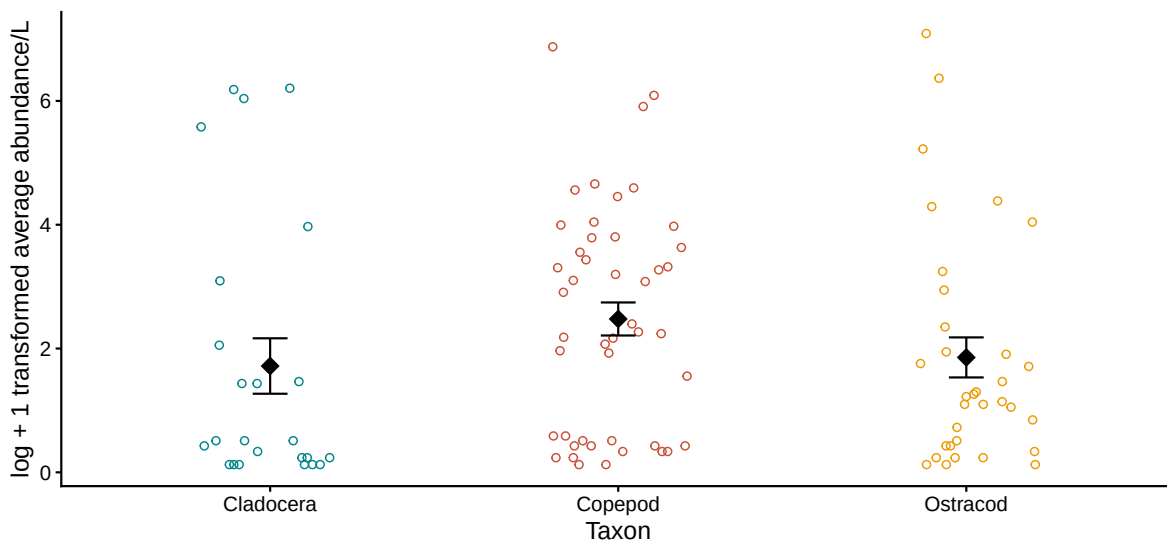
(JUNE) - Describe temporal and spatial coverage of dataset - Reference monitoring report or NCOS-related publications on data collection - Describe frequency of data collection - Describe statistical analysis or descriptive statistics (linked to your question) ->

Results

(NATALIE) - Discuss findings clearly (patterns, trends, etc.) - Connect results to your research questions and figures - Present figures in an order that tells a clear ecological story

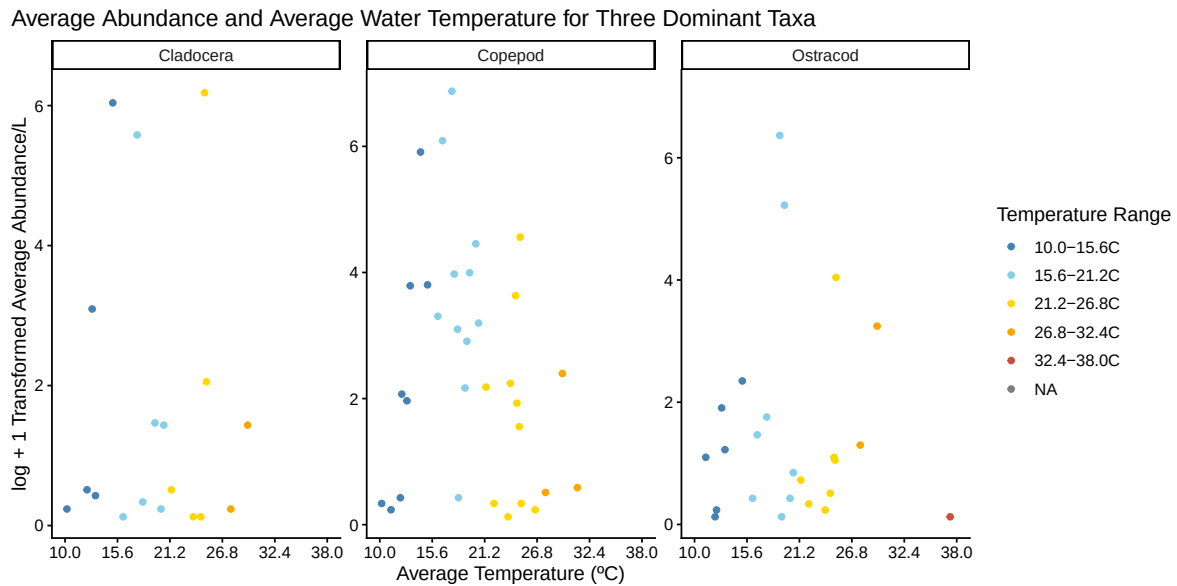
Figure 1 - What is the average abundance distribution of the three dominant taxa?

Mean and Distribution of Average Abundance for Three Dominant Taxa



- First we analyzed the variance of average abundance/L across our three taxa in our scope. We found that Copepod was the most abundant at NCOS, followed by Ostracod, then slightly behind by Cladocera. Cladocera seemed to have the widest distribution of average abundance/L, whereas Copepod had the smallest and more consistent distribution across various sample points, and Ostracod had a moderate sized distribution. This figure supports our research question as it visualized specifically *how* abundant each taxa is and how spread out their data points were across all sites and temperatures.

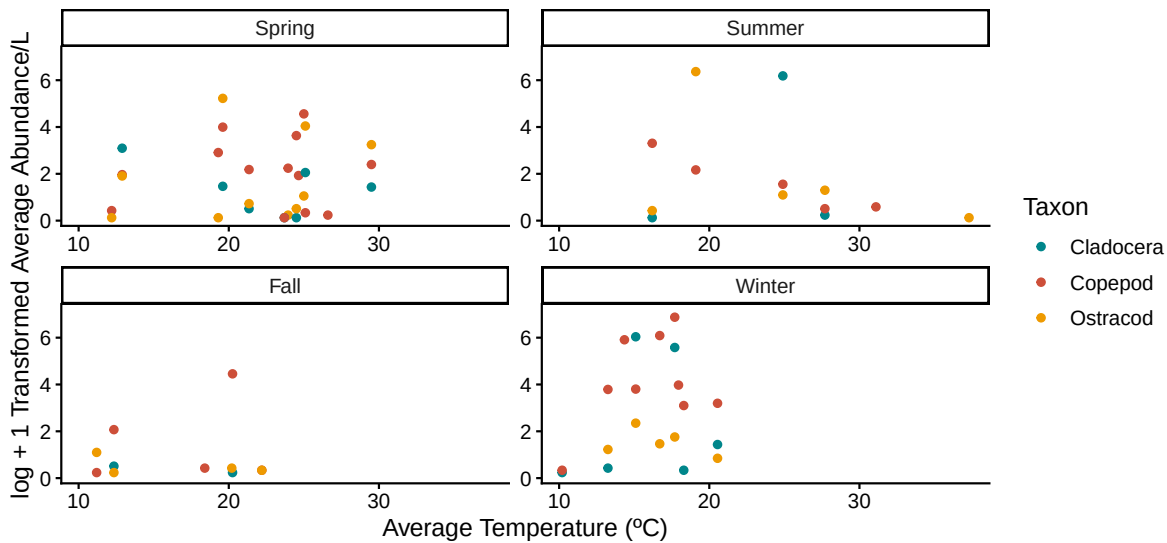
Figure 2 - How does abundance vary across temperature ranges for each taxon?



- After understanding specifically *how* abundant each taxa was across all sites, we created Figure 2 based on average abundance and average water temperature per sample. Average temperatures varied from 0-38°C, thus we split up the whole temperature range equally into 5 color coded ranges: **dark blue** (10.0-15.6°C), **light blue** (15.6-21.2°C), **yellow** (21.2-26.8°C), **orange** (26.8-32.4°C), and **red** (32.4-38.0°C). Cladocera was most abundant in a temperature range between 10.0-21.2°C, Copepod had the widest abundance range of 10.0-26.8°C, and Ostracod was most abundant between 15.6°-26.8°C. This figure supports our research question as it explains in which specific range of temperature was there the most average abundance of taxon/L.

Figure 3 - What is the strength of the relationship between average water temperature and average invertebrate abundance across by season?

Average Abundance and Average Water Temperature by Season for Three Dominant Taxa



- It was clear from Figure 2 that aquatic invertebrates tended to be more abundant from an average water temperature from 10.0-26.8°C. Thus, we further explored *when* exactly in the year this water temperature tended to occur. Figure 3 shows average abundance/L by average water temperature (°C) for each of the three taxa per season, between 2020-2024. Winter stood out to have the lowest average water temperatures between 10.0-20°C, thus we concluded that average aquatic invertebrate abundance is most likely highest during the Winter at NCOS. This figure connects to our research question as it answers the correlation between water temperature, aquatic abundance during a specific time range.

Discussion

Connecting Results to NCOS Reports/Publications

The results align with findings from Stiles & Kracha, whose poster on a restored California wetland documented shifts in invertebrate assemblages as the system matured over three years (Stiles and Kracha 2023). Given that NCOS restoration began in 2017 and sampling commenced in 2018, the data collected likely reflect an ecosystem still establishing ecological equilibrium. The winter abundance spike observed is consistent with Stiles & Kracha's finding that seasonal rainfall reduces salinity stress, driving fluctuations in invertebrate community composition (Stiles and Kracha 2023).

Connecting Results to Other Ecosystems

The temperature-abundance relationship observed at NCOS finds support in studies conducted across comparable wetland systems. Marchetti et al. found that invertebrate abundance, richness, and diversity increased with wetland age across restored Sacramento Valley wetlands, with Ostracoda — one of the three taxa monitored in this study — comprising over 20% of all samples (Marchetti, Garr, and Smith 2010). Given that NCOS is a relatively young restoration site, the abundance patterns documented here likely reflect an early stage of that same successional trajectory. Suren et al. similarly demonstrated that invertebrate communities recolonized successfully following hydrological restoration with minimal intervention, suggesting that restored wetlands across geographic contexts follow comparable recovery dynamics (Suren, Lambert, and Sorrell 2011).

Challenges in Data Collection

Obtaining representative water samples was complicated by particulate matter — including mud, sticks, and organic debris — that required filtration and likely introduced variability in usable sample volume across collection events. Some sampling locations, such as areas beneath bridges or in seasonally waterlogged margins, posed physical access challenges that may have limited consistency across seasons.

Other Variables That Would Influence Observed Patterns

Salinity is the most directly relevant confounding variable, as winter rainfall reduces salinity independently of temperature, meaning some of the abundance patterns observed may partly reflect salinity dynamics rather than thermal effects alone; salinity is typically measured using a conductivity meter or refractometer at the time of collection. Dissolved oxygen (DO) is an additional variable of interest, given that warmer water holds less oxygen and could suppress invertebrate abundance at elevated temperatures independently of direct thermal stress; DO is measured via probe concurrent with sampling. Local vegetation cover and canopy shading may also introduce spatial variability, as shaded microsites can register cooler temperatures independent of seasonal conditions.

Temporal Changes and Dataset Limitations

Seasonal sampling captures broad temporal trends but cannot account for finer-scale environmental variation; a single sample per season represents a snapshot that may not reflect conditions experienced by invertebrate populations over preceding weeks. Since invertebrate community structure responds to accumulated conditions rather than instantaneous measurements, short-term fluctuations between sampling events go undetected. Without continuously deployed temperature loggers at each of the 12 Devereux Slough sites, it remains difficult

to assess whether recorded temperatures are representative of the broader inter-sampling period, and multi-year data would be needed to distinguish restoration-driven trajectories from recurring seasonal cycles.

References

- Marchetti, Michael P., Mariah Garr, and Adam N. H. Smith. 2010. “Evaluating Wetland Restoration Success Using Aquatic Macroinvertebrate Assemblages in the Sacramento Valley, California.” *Restoration Ecology* 18: 457–66. <https://doi.org/10.1111/j.1526-100X.2008.00468.x>.
- Stiles, L., and C. Kracha. 2023. “Assessing Aquatic Invertebrate Establishment in a Restored Wetland over Time.” UC Santa Barbara: Cheadle Center for Biodiversity; Ecological Restoration. <https://escholarship.org/uc/item/00t2k2t7>.
- Suren, Alastair M., Paul Lambert, and Brian K. Sorrell. 2011. “The Impact of Hydrological Restoration on Benthic Aquatic Invertebrate Communities in a New Zealand Wetland.” *Restoration Ecology* 19: 747–57. <https://doi.org/10.1111/j.1526-100X.2010.00723.x>.